

Mathematics A

Unit 1 • Chapter OL-T49 • Expertise Q16+ Classified

Cross-session • chapter-organised candidate

5 classified items • 26 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

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Contents

Unit 1 • Expertise • Unit 1 • Chapter OL-T49 • Expertise Q16+ • 5 items • 26 marks

Chapter	Items	Marks	Starts
01. OL-T49 Sine, Cosine Rule & Area of Triangles	5	26	4

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Sine, Cosine Rule & Area of Triangles

Unit 1 • Expertise • 26 marks • 5 item(s)

June 2025 | 4MA1-1H Q20 | 5 marks

Use two non-right triangles sharing a diagonal

June 2023 | 4MA1-2H Q22 | 5 marks

Combine non-right trigonometry with circle or elevation geometry

June 2022 | 4MA1-1H Q18 | 5 marks

Find an acute angle using the sine rule

January 2022 | 4MA1-2H Q23 | 6 marks

Resolve the ambiguous sine-rule angle

January 2023 | 1H Q16 | 5 marks

Find a side using the sine rule

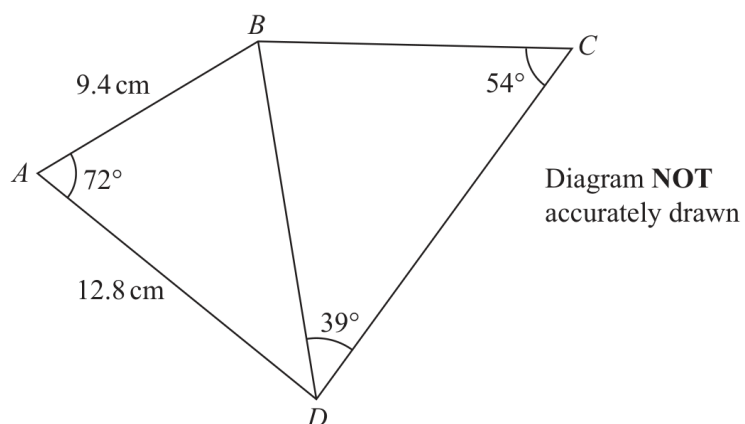
Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

4MA1-1H Q20

5 marks

20



Work out the length of BC
Give your answer correct to 3 significant figures.

..... cm

(Total for Question 20 is 5 marks)

WORKING SPACE

4MA1-2H Q22

5 marks

22 The diagram shows a triangle ABC and a flagpole BF

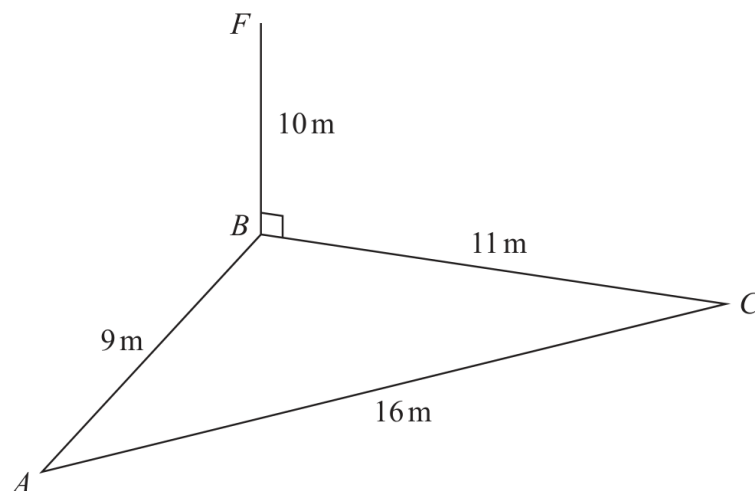


Diagram **NOT**
accurately drawn

A , B and C are points on horizontal ground.

BF is vertical.

$$AB = 9 \text{ m} \quad BC = 11 \text{ m} \quad AC = 16 \text{ m} \quad BF = 10 \text{ m}$$

D is the point on AC such that angle $BDC = 90^\circ$

Work out the size of the angle of elevation of the point F from the point D
Give your answer correct to one decimal place.

○

Continue your method**5 marks**

WORKING SPACE

4MA1-1H Q18 Q18

5 marks

18 Here is triangle ABC

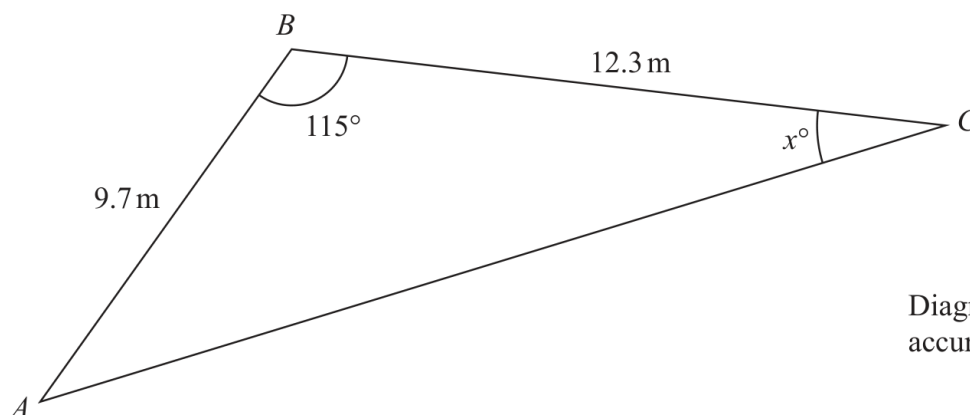


Diagram **NOT**
accurately drawn

Work out the value of x
Give your answer correct to 3 significant figures.

$x =$

Continue your method**5 marks**

WORKING SPACE

4MA1-2H Q23

6 marks

23 The diagram shows triangle PQR

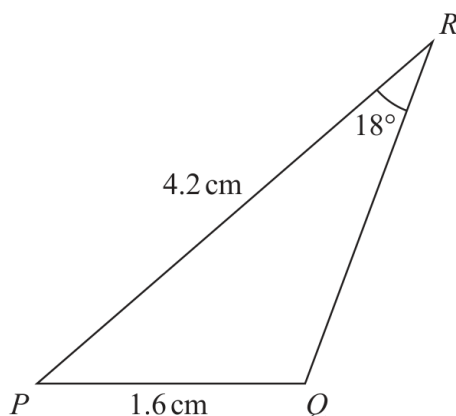


Diagram **NOT**
accurately drawn

$$PQ = 1.6 \text{ cm}$$

$$PR = 4.2 \text{ cm}$$

$$\text{Angle } PRQ = 18^\circ$$

Given that angle PQR is obtuse,

work out the area of triangle PQR

Give your answer correct to 3 significant figures.

..... cm^2

Continue your method**6 marks**

WORKING SPACE

- 16 Here is a shape formed from two triangles ABC and CDE
 ACD and BCE are straight lines.

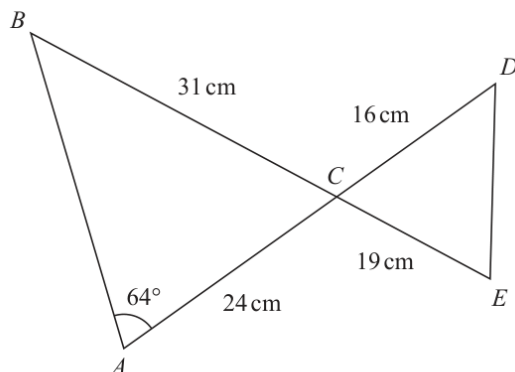


Diagram **NOT**
 accurately drawn

$$AC = 24 \text{ cm} \quad BC = 31 \text{ cm} \quad CE = 19 \text{ cm} \quad CD = 16 \text{ cm}$$

$$\text{Angle } BAC = 64^\circ$$

Work out the length of DE

Give your answer correct to 3 significant figures.



YOUR WORKING

January 2023 | 1H | Question 16 | 5 marks

Chapter OL-T49: Sine, Cosine Rule & Area of Triangles

.....cm

(Total for Question 16 is 5 marks)

YOUR WORKING